Manual Deployment Architecture

```
Internet / Users
      |
      v
http://10.43.169.43
      |
      v
Nginx (Port 80)
      |
      v
Gunicorn (127.0.0.1:8000)
      |
      v
Flask App
      |
      v
MongoDB (27017)
```

Layer 1 - Application

```
Commands:
cd /home/ute/Manual-deployment-github/Manual_app_deployment
source venv/bin/activate
python app.py

Checks:
curl http://127.0.0.1:5000
curl http://127.0.0.1:5000/health

Purpose: Verify application works before deployment.
```

Layer 2 - Database

```
Commands:
sudo systemctl status mongod
ss -tulpn | grep 27017
mongosh

Purpose: Ensure the application can communicate with MongoDB.
```

Layer 3 - Production Application Server (Gunicorn)

```
Install:
pip install gunicorn

Run:
gunicorn --bind 0.0.0.0:8000 app:app

Checks:
curl http://127.0.0.1:8000/health

Purpose: Production-grade application server with multiple workers and better reliability.
```

Layer 4 - Process Management (systemd)

```
Purpose:
- Auto Start
- Auto Restart
- Background Execution
- Centralized Logging

Important Commands:
sudo systemctl daemon-reload
sudo systemctl start taskmanager
sudo systemctl stop taskmanager
sudo systemctl restart taskmanager
sudo systemctl status taskmanager
sudo journalctl -u taskmanager -f
```

Layer 5 - Reverse Proxy (Nginx)

```
Purpose: Users access port 80 instead of port 8000.

Flow:
Browser -> Nginx:80 -> Gunicorn:8000

Important Commands:
sudo nginx -t
sudo systemctl restart nginx
sudo systemctl status nginx
```

Layer 6 - Security

```
Public Ports:
22 SSH
80 HTTP
443 HTTPS

Private Ports:
8000 Gunicorn
27017 MongoDB

Rule: Never expose Gunicorn or MongoDB directly to users.
```

Layer 7 - Troubleshooting Flow

```
Step 1: systemctl status nginx
Step 2: systemctl status taskmanager
Step 3: journalctl -u taskmanager -f
Step 4: curl http://127.0.0.1:8000/health
Step 5: systemctl status mongod
```

Complete Deployment Procedure

```
Phase 1: Verify Flask Application
1. Activate virtual environment.
2. Run python app.py.
3. Verify using curl http://127.0.0.1:5000/health.

Phase 2: Verify MongoDB
1. systemctl status mongod
2. Verify port 27017.
3. Login using mongosh.

Phase 3: Run Gunicorn
1. pip install gunicorn
2. gunicorn --bind 0.0.0.0:8000 app:app
3. Test health endpoint.

Phase 4: Create systemd Service
Service file path:
/etc/systemd/system/taskmanager.service

ExecStart:
/home/ute/Manual-deployment-github/Manual_app_deployment/venv/bin/gunicorn --workers 3 --bind 127.0.

Commands:
sudo systemctl daemon-reload
sudo systemctl start taskmanager
sudo systemctl enable taskmanager
sudo systemctl status taskmanager

Phase 5: Configure Nginx
File:
/etc/nginx/conf.d/taskmanager.conf

Server listens on port 80 and proxies traffic to 127.0.0.1:8000.

Commands:
sudo nginx -t
sudo systemctl restart nginx

Phase 6: Configure Firewall
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload

Phase 7: Validation
curl http://127.0.0.1:8000/health
curl http://127.0.0.1/health
Open browser: http://10.43.169.43

Final Learning:
Application -> Database -> Gunicorn -> systemd -> Nginx -> Firewall -> Validation
Build and test one layer at a time.
```